Lecture 38: Deep Learning (Part-3) — Practice Problems Suite (Hinglish)

Yeh problem suite PyTorch mechanics, ANN regression, multi-class classification, preprocessing pipelines, model checkpointing, aur evaluation metrics par based hai.

Problem 1: Regression vs Multi-Class Classification Architecture

ANN regression aur multi-class classification ($C = 7$) ke output layer design aur loss function ka comparison karein. Regression mein 1 output neuron bina kisi activation ke kyun hota hai, jabki classification mein C neurons hote hain aur PyTorch model ke andar explicit Softmax kyun nahi lagaya jata?

Problem 2: Feature Standardization Ki Zaroorat

Deep neural networks ko train karte waqt `StandardScaler` lagana mathematically kyun zaroori hai? Agar features ka scale alag-alag ho (`Area` $\approx 500,000$ aur `Eccentricity` ≈ 0.7), toh loss surface aur gradient descent par kya asar padta hai?

Problem 3: Data Leakage Feature Scaling Mein

Agar hum `train_test_split` karne se pehle poore dataset par `scaler.fit_transform(X)` chala dein, toh isko Data Leakage kyun kehte hain? `fit_transform` aur `transform` use karne ka sahi tareeqa kya hai?

Problem 4: PyTorch Data Pipeline Mechanics

`TensorDataset` aur `DataLoader` ka role kya hota hai? Hum poore dataset tensor ko seedhe `model(X)` mein kyun nahi pass karte? Mini-batching ($B = 32$) ka memory aur optimization benefit kya hai?

Problem 5: Tensor Data Types (`torch.float32` vs `torch.long`)

PyTorch mein regression ke targets ke liye `torch.float32` aur multi-class classification ke targets ke liye `torch.long` (`int64`) kyun zaroori hota hai? Agar classification target mein float pass kar diya jaye toh kya error aayega?

Problem 6: Manual Parameter Count Calculation

Dono models ke learnable parameters (weights + biases) ka manual calculation karein:

1. Regression model: Input $4 \rightarrow$ Linear(64) $\rightarrow$ Linear(32) $\rightarrow$ Linear(1).
2. Classification model: Input $34 \rightarrow$ Linear(64) $\rightarrow$ Linear(64) $\rightarrow$ Linear(7).

Problem 7: Softmax aur `nn.CrossEntropyLoss` Ka Common Trap

PyTorch classification model ke aakhri layer mein `nn.Softmax()` lagana ek serious bug kyun mana jata hai jab loss `nn.CrossEntropyLoss()` ho? Log-Sum-Exp trick kya hoti hai?

Problem 8: PyTorch Optimization Ke 4 Pillars

PyTorch training loop ke 4 core steps explain karein:

1. `optimizer.zero_grad()`
2. `outputs = model(inputs)`
3. `loss.backward()`
4. `optimizer.step()` Agar `optimizer.zero_grad()` ko comment kar diya jaye toh model par kya impact padega?

Problem 9: `model.state_dict()` vs Full Model Save Karna

`model.state_dict()` kya hota hai? Model ko `torch.save(model.state_dict(), path)` se save karna pure object `torch.save(model, path)` se behtar kyun hai?

Problem 10: Model Checkpointing aur Overfitting Se Bachav

Model checkpointing ka logic kya hai? Best model select karne ke liye training loss ke bajaye validation loss kyun dekha jata hai? `best_val_loss = float('inf')` kyun set kiya jata hai?

Problem 11: Evaluation Guardrails (`model.eval()` aur `torch.no_grad()`)

`model.eval()` aur `with torch.no_grad():` mein kya farq hai? Kya `model.eval()` gradient tracking band karta hai? Kya `torch.no_grad()` dropout ko disable karta hai?

Problem 12: Accuracy Calculation Mein Dynamic Batch Size (`yb.size(0)`)

Evaluation loop mein `total += yb.size(0)` likhna sahi kyun hai aur `total += 32` likhna ek mathematical bug kyun create karta hai jab dataset size 180 ho?

Problem 13: Regression Metrics Ka Interpretation

Power plant regression problem mein model ne test MSE 17.81 aur R^2 score 0.937 achieve kiya:

1. Test RMSE calculate karein aur electricity power (MW) ke context mein iska matlab samjhayein.
2. $R^2 = 0.937$ ka kya statistical meaning hai?

Problem 14: Logits vs Probabilities Par Argmax Ki Equivalence

Classification testing mein hum `_, predicted = torch.max(outputs, 1)` raw logits par lagate hain. Prove karein ki raw logits par argmax lena aur Softmax probabilities par argmax lena bilkul same result deta hai.

Problem 15: PCA + ANN Dimensionality Reduction Challenge

Date fruit dataset ke 34 features par PCA kaise apply kiya jayega? Agar 95% cumulative variance preserve karne ke liye k principal components chune jayein, toh PyTorch architecture mein kya badlav aayega?